

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")
```

```
In [3]: df = pd.read_csv("titanic.csv")
df.head()
```

```
Out[3]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0

Observation: Dataset loaded successfully. First 5 rows displayed

```
In [4]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age         714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

```

Observation:

Check total rows and columns. Identify missing values.

In [5]: `df.describe()`

Out[5]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.200000
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693000
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910000
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.450000
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.320000

Observation:

Average age Average fare Minimum and maximum values

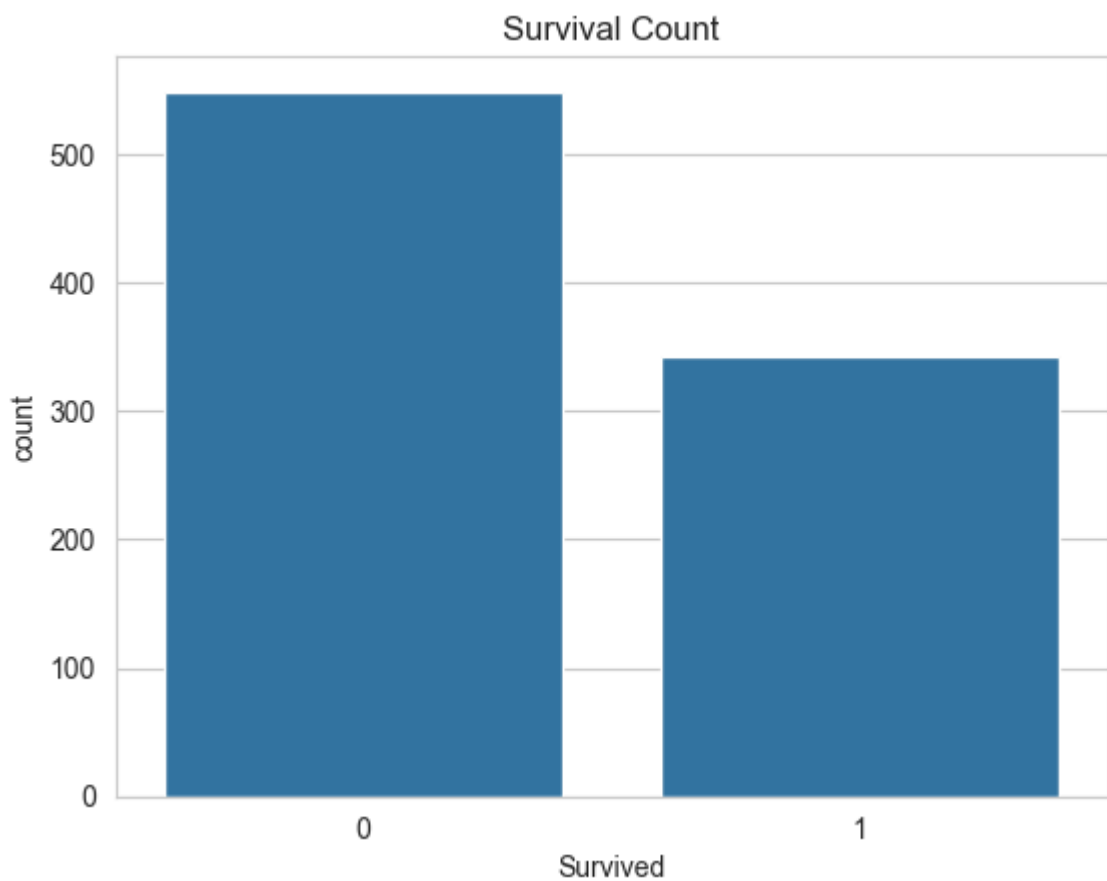
In [6]: `df.isnull().sum()`

```
Out[6]: PassengerId    0
        Survived      0
        Pclass       0
        Name         0
        Sex          0
        Age         177
        SibSp        0
        Parch        0
        Ticket       0
        Fare         0
        Cabin       687
        Embarked     2
        dtype: int64
```

Observation:

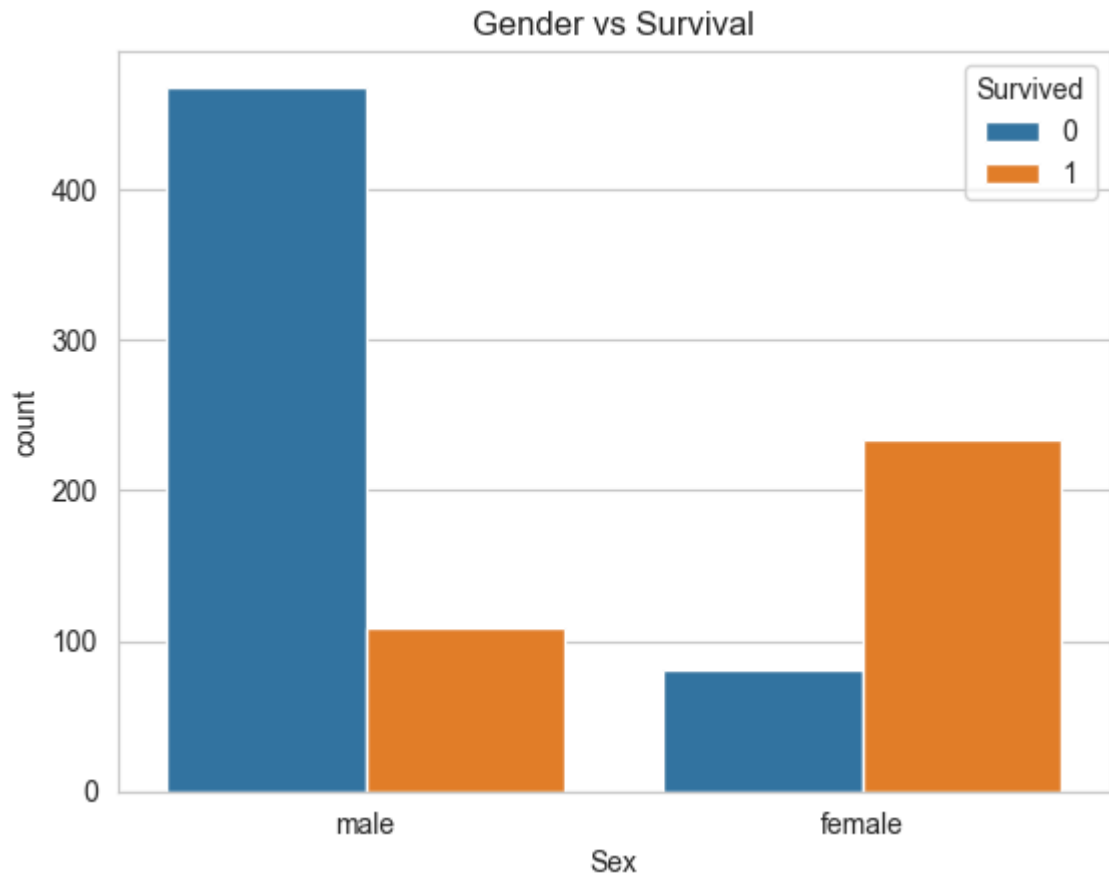
Age contains missing values. Cabin contains many missing values.

```
In [7]: sns.countplot(x='Survived', data=df)
        plt.title("Survival Count")
        plt.show()
```



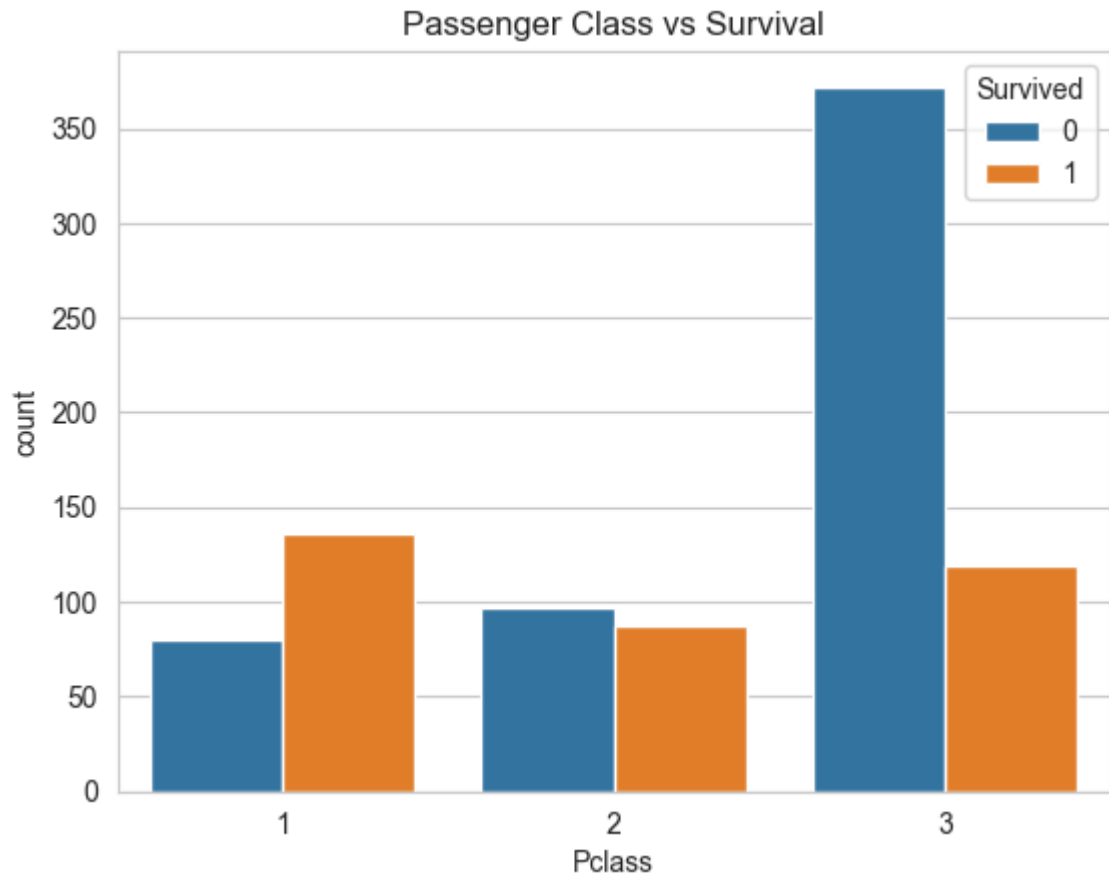
More passengers died than survived.

```
In [8]: sns.countplot(x='Sex', hue='Survived', data=df)
        plt.title("Gender vs Survival")
        plt.show()
```



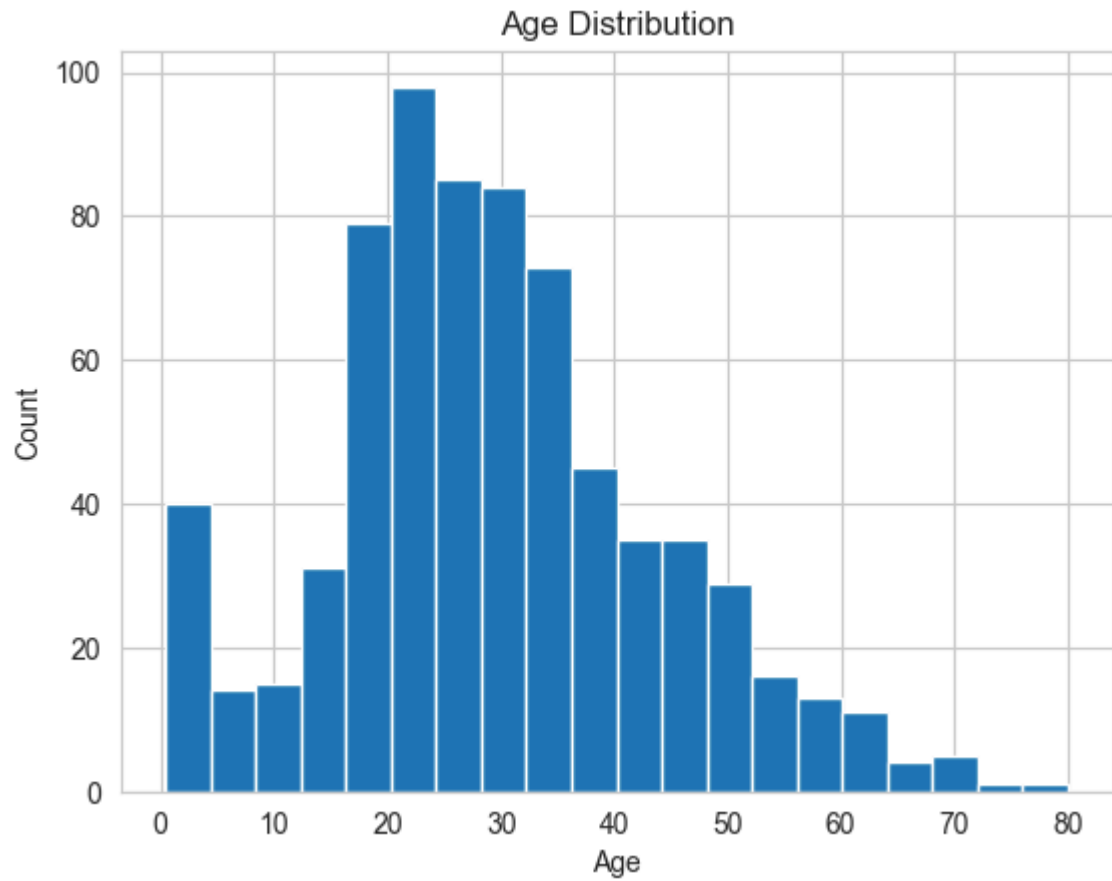
Females had higher survival rate than males.

```
In [9]: sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title("Passenger Class vs Survival")
plt.show()
```



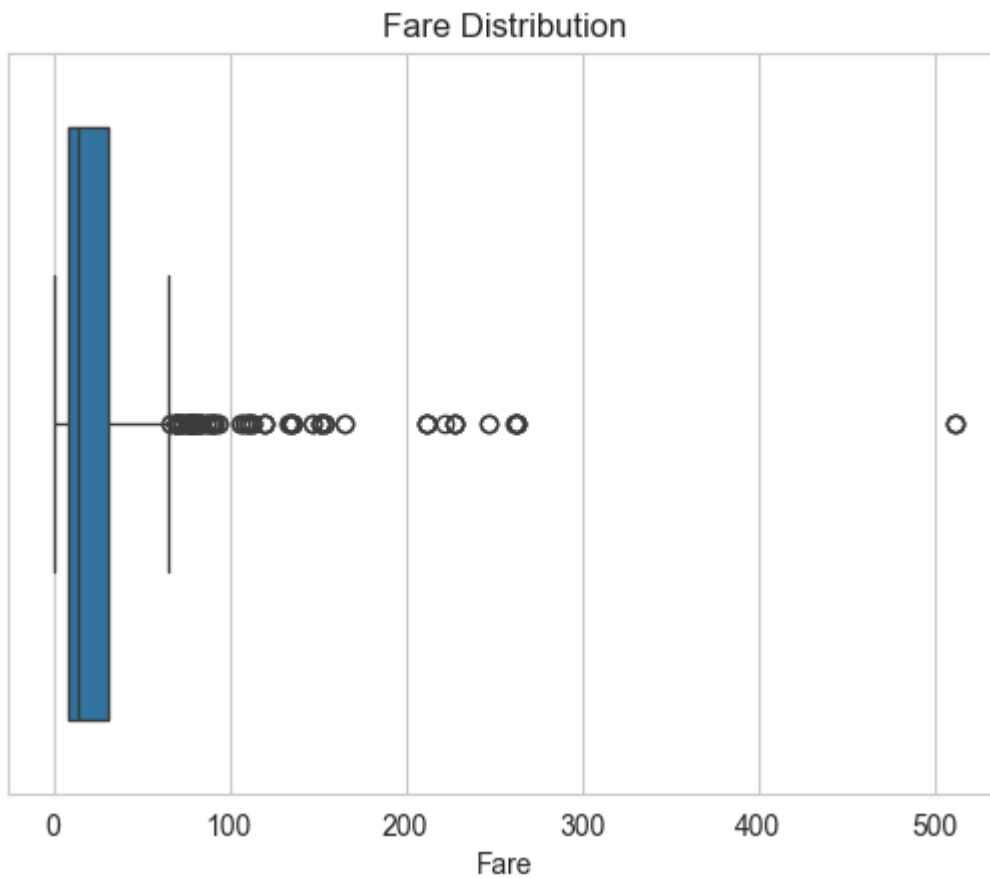
First-class passengers survived more often.

```
In [10]: plt.hist(df['Age'].dropna(), bins=20)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
```



Most passengers were between 20 and 40 years old.

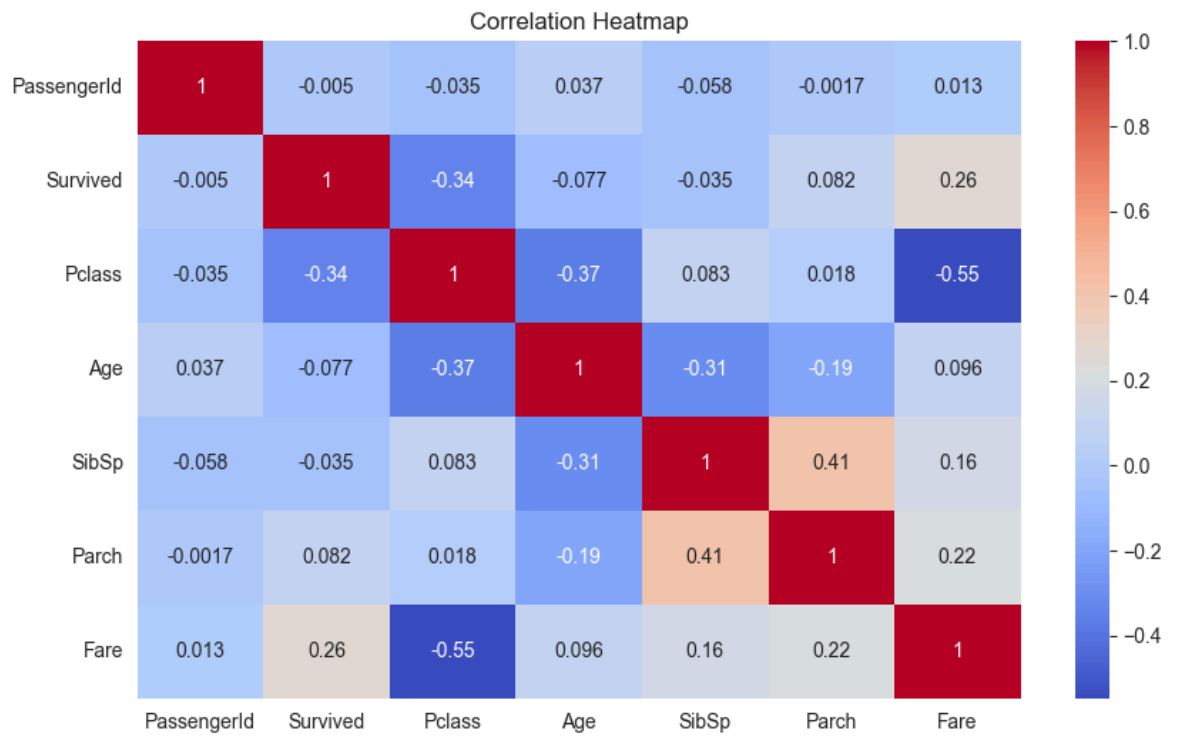
```
In [11]: sns.boxplot(x=df['Fare'])  
plt.title("Fare Distribution")  
plt.show()
```



Several outliers exist in fare values.

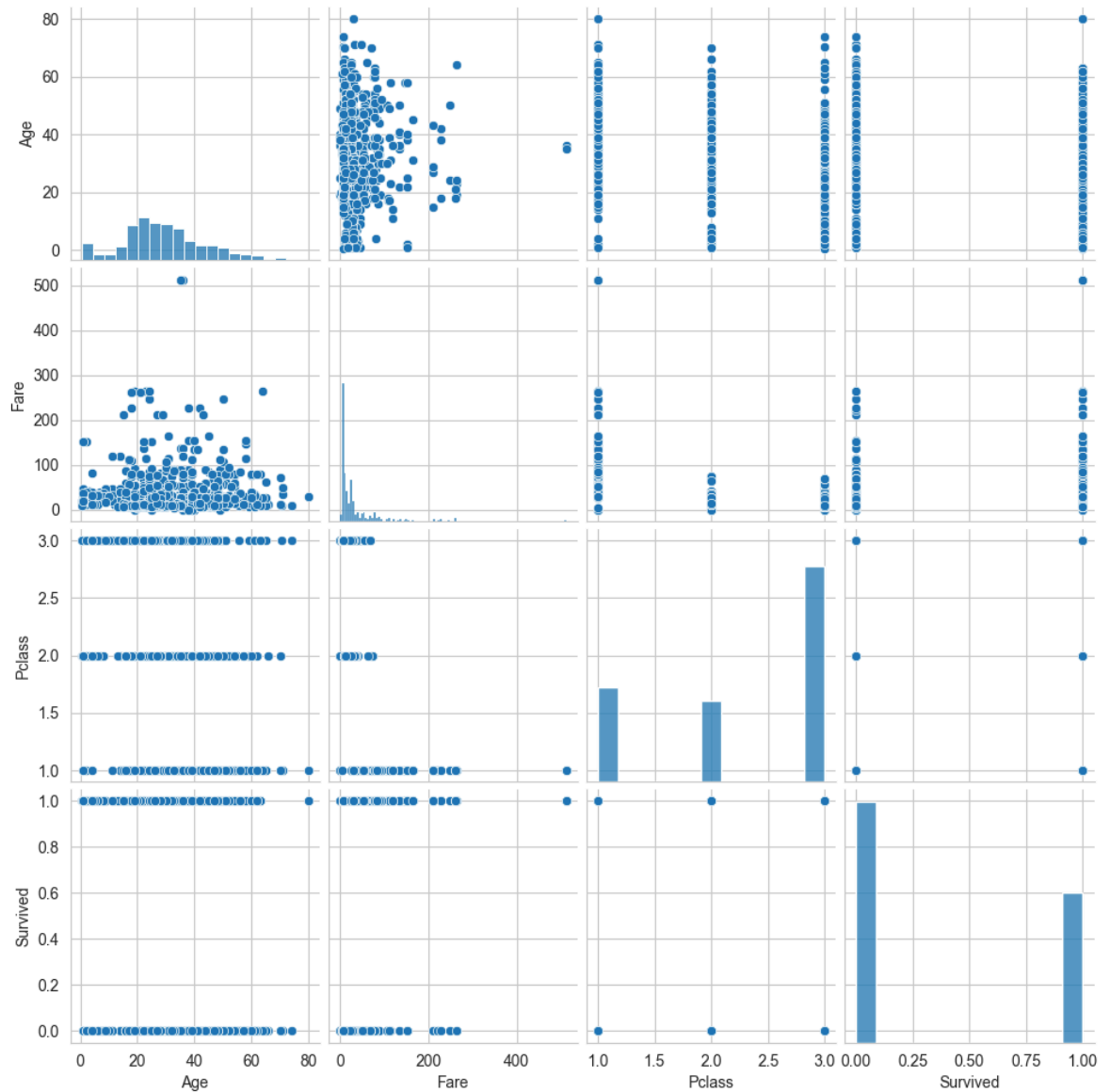
```
In [12]: numeric_df = df.select_dtypes(include=np.number)

plt.figure(figsize=(10,6))
sns.heatmap(numeric_df.corr(),
            annot=True,
            cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



Fare and Pclass show noticeable correlation.

```
In [16]: sns.pairplot(df[['Age', 'Fare', 'Pclass', 'Survived']])  
plt.show()
```

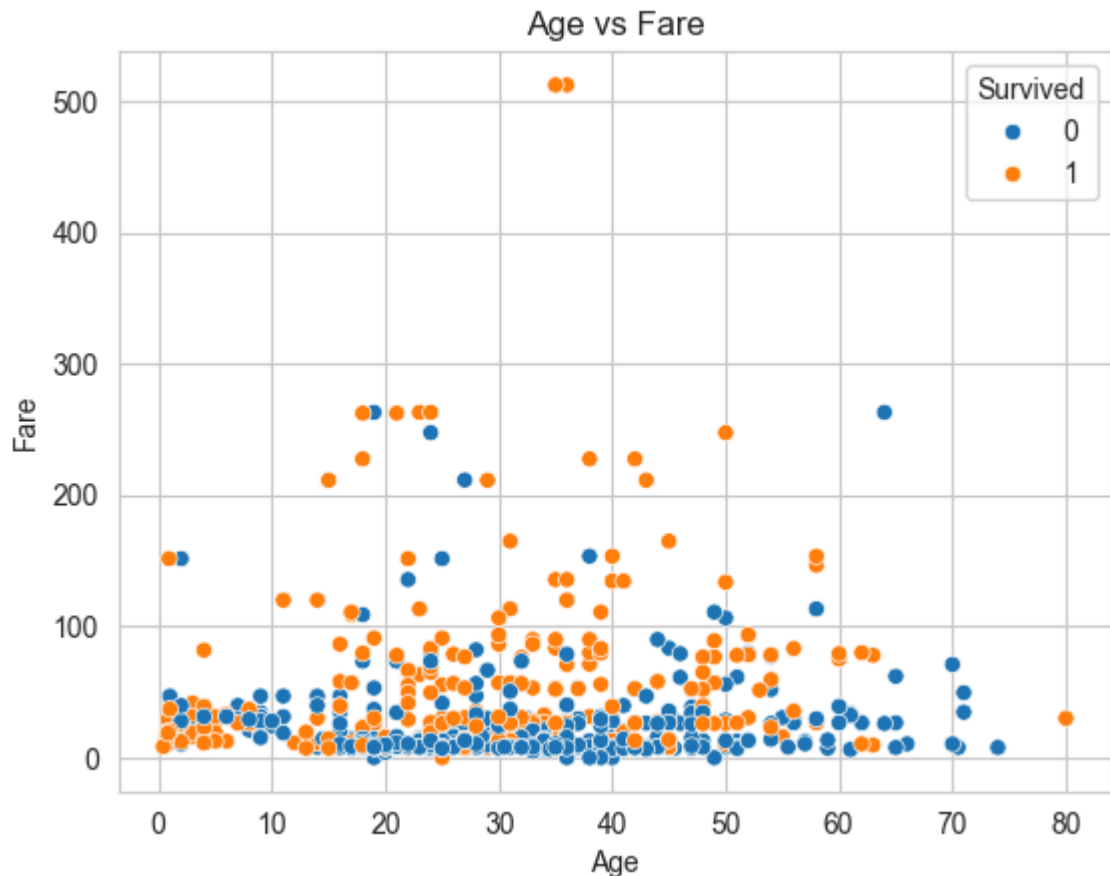



Observation:

Shows relationships among Age, Fare, Passenger Class, and Survival.

```
In [14]: sns.scatterplot(x='Age',
                        y='Fare',
                        hue='Survived',
                        data=df)

plt.title("Age vs Fare")
plt.show()
```



In []: Observation:

Higher fares generally belong to higher passenger classes.

```
In [15]: print(df['Sex'].value_counts())
print(df['Pclass'].value_counts())
```

```
Sex
male      577
female    314
Name: count, dtype: int64
Pclass
3      491
1      216
2      184
Name: count, dtype: int64
```

Observation:

Male passengers were more than female passengers. Most passengers belonged to 3rd class.

SUMMARY OF FINDINGS

1. More passengers died than survived.
2. Female passengers had a higher survival rate.
3. First-class passengers survived more often.
4. Most passengers were aged between 20 and 40.
5. Fare contains several outliers.
6. Passenger class influences survival.

7. Strong relationships observed between Fare and Pclass.